

unstable, unreliable and not environmentally-friendly. The team has come up with a method to use fluid power technology for VTOL AAV.

This new technology is an inexpensive, recyclable hydraulic propulsion system for the multi-rotor VTOL aircraft. The propulsion system uses hydrostatic transmission, a lighter weight and more reliable option, to distribute engine power throughout the rotors, providing thrust for the aircraft and allowing the rotors to each spin at different speeds.

Speed of each motor can be controlled individually with faster response by the flight controller or human operator, and can run at constant speeds, extending the engine lifetime. This provides both aerodynamic lift and attitude control, eliminating the need for an additional moving control surface or weight shifting device, and resulting in a more stable flight and more useful load.

Washable, wearable battery-like devices that could be woven directly into clothes

Wearable electronic components incorporated directly into fabrics have been developed by researchers at University of Cambridge. The devices could be used for flexible circuits, healthcare monitoring, energy conversion and other applications.



Washable, wearable batteries based on cheap, safe and environmentally-friendly inks and woven directly into fabrics (Credit: University of Cambridge)

The Cambridge researchers, working in collaboration with colleagues at Jiangnan University in China, have shown how graphene—a two-dimensional form of carbon—and other related materials can be directly incorporated into fabrics to produce charge storage elements such as capacitors, paving the way for textile-based power supplies, which are washable, flexible and comfortable to wear.

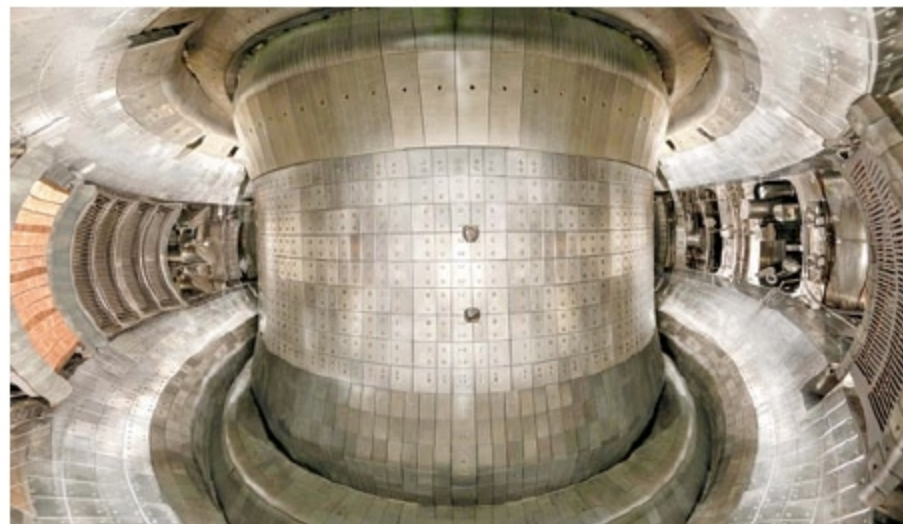
The research shows that graphene inks can be used in tex-

tiles to store electrical charge and release it when required. The new textile electronic devices are based on low-cost, sustainable and scalable dyeing of polyester fabric. Inks are produced by standard solution processing techniques.

Building on previous work by the same team, researchers designed inks that can be directly coated onto a polyester fabric in a simple dyeing process. Versatility of the process allows various types of electronic components to be incorporated into the fabric.

China's pursuit for clean, limitless energy heats up

A ground-breaking fusion reactor built by Chinese scientists is underscoring Beijing's determination to be at the core of clean energy technology as it eyes a fully-functioning plant by 2050. Sometimes called an artificial Sun for the sheer heat and power it produces, the doughnut-shaped Experimental Advanced Superconducting Tokamak (EAST), which juts out on a spit of land into a lake in eastern Anhui province, has notched up a succession of firsts.



A vacuum vessel in Experimental Advanced Superconducting Tokamak device in Hefei, China (Credit: AFP-JIJI)

In November 2018, it became the first facility in the world to generate 100 million degrees Celsius—six times hotter than the Sun's core. Such mind-boggling temperatures are critical to achieving sustainable nuclear fusion reactions, which promise an inexhaustible energy source.

EAST's main reactor stands within a concrete structure, with pipes and cables spread outward like spokes that connect to a jumble of sensors and other equipment encircling the core.

China is also aiming to build a separate fusion reactor that could begin generating commercially-viable fusion power by mid-century. Around six billion yuan (US\$ 890 million) has been promised for the ambitious project, according to *The Japan Times*.

EAST is part of International Thermonuclear Experimental Reactor (ITER) project, which seeks to prove the feasibility of fusion power. Funded and run by the European Union, India, Japan, China, Russia, South Korea and the US, the project's centre-piece will be a giant cylindrical fusion device, called a tokamak.